



INSPIRE FORENSICS

(Recognized by the Indian Society of Forensic Science)

CUET-UG 2026

SAMPLE PAPER

A National Level Socio-Forensic Voluntary Student-Led Educational Firm

Biology/ Biological Studies/ Biotechnology/Biochemistry (304)

Q1. Which hormone shows negative feedback with ACTH? [NEET 2022 PYQ]

- A. Adrenaline
- B. Aldosterone
- C. Cortisol
- D. Thyroxine

Q2. Which taxonomic hierarchy shows the maximum genetic similarity? [CUET 2023 PYQ]

- A. Family
- B. Genus
- C. Species
- D. Order

Q3. Which base pair has THREE hydrogen bonds? [NEET 2018 PYQ]

- A) A–T
- B) A–U
- C) G–C
- D) C–T

Q4. Which forensic sample provides mitochondrial DNA when nuclear DNA is absent?

- A. Blood plasma
- B. Hair shaft
- C. Sweat
- D. Saliva

Q5. Which histone protein is NOT part of nucleosome core? [NEET 2018 PYQ]

- A. H2A**
- B. H2B**
- C. H3**
- D. H1**

Q6. Which structure filters blood in nephron? [NEET 2021 PYQ]

- A. Loop of Henle**
- B. Bowman's capsule**
- C. DCT**
- D. Collecting duct**

Q7. Which mutation produces maximum change in protein sequence? [NEET 2019 PYQ]

- A. Silent**
- B. Missense**
- C. Nonsense**
- D. Frameshift**



Q8. Which organ has highest regenerative capacity? [NEET 2018 PYQ]

- A. Brain**
- B. Kidney**
- C. Liver**
- D. Heart**

Q9. Which enzyme synthesizes RNA using DNA template? [CUET 2022 PYQ]

- A. DNA polymerase**
- B. RNA polymerase**
- C. Reverse transcriptase**
- D. Ligase**

Q10. Which antibiotic inhibits cell wall synthesis? [NEET 2020 PYQ]

- A. Streptomycin**
- B. Penicillin**
- C. Tetracycline**
- D. Chloramphenicol**

Q11. Affected father → all daughters affected → no sons affected indicates: [NEET 2022 PYQ]

- A. Autosomal recessive**
- B. Autosomal dominant**
- C. X-linked dominant**
- D. Y-linked**

Q12. Which blotting technique detects DNA? [NEET 2020 PYQ]

- A. Western**
- B. Northern**
- C. Southern**
- D. Eastern**



Q13. Which enzyme removes RNA primer in eukaryotes?

- A. DNA Pol I**
- B. RNase H**
- C. Ligase**
- D. Primase**

Q14. Which gas has highest affinity for hemoglobin? [NEET 2020 PYQ]

- A. O₂**
- B. CO₂**
- C. CO**
- D. N₂**

Q15. DNA fingerprinting is based on polymorphism in: [CUET 2023 PYQ]

- A. Exons**
- B. Introns**
- C. STRs**
- D. rRNA**

Q16. Which structure degenerates if fertilization does not occur? [NEET 2020 PYQ]

- A. Corpus luteum**
- B. Graafian follicle**
- C. Zona pellucida**
- D. Endometrium**

Q17. Which population shows Hardy–Weinberg equilibrium? [NEET 2021 PYQ]

- A. Migrating**
- B. Mutating**
- C. Random mating**
- D. Small population**



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Q18. Which leukocyte is MOST involved in allergic reactions? [NEET 2021 PYQ]

- A. Neutrophil**
- B. Basophil**
- C. Monocyte**
- D. Lymphocyte**

Q19. Which enzyme prevents DNA overwinding during replication? [NEET 2020 PYQ]

- A. Helicase**
- B. Ligase**
- C. Gyrase**
- D. Primase**

Q20. Which immunoglobulin crosses placenta?

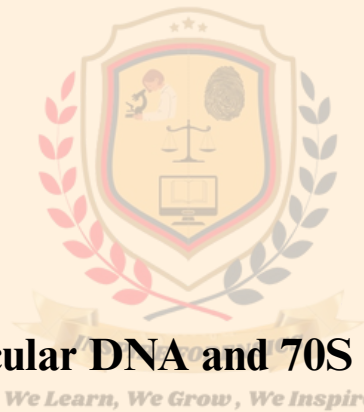
- A. IgM**
- B. IgA**
- C. IgE**
- D. IgG**

Q21. Which codon acts as START codon? [CUET 2022 PYQ]

- A. UAA**
- B. UAG**
- C. AUG**
- D. UGA**

Q22. Which chromosome combination determines male sex in humans? [CUET 2022 PYQ]

- A. XX**
- B. XO**
- C. XY**
- D. YY**



Q23. Which organelle has circular DNA and 70S ribosomes? [CUET 2022 PYQ]

- A. Nucleus**
- B. Lysosome**
- C. Mitochondria**
- D. Golgi body**

Q24. Which taxonomic aid is MOST useful for forensic entomology in estimating PMI?

- A. Herbarium**
- B. Monograph**
- C. Dichotomous key**
- D. Botanical Garden**

Q25. Which process creates genetic recombination? [NEET 2019 PYQ]

- A. Translation**
- B. Replication**
- C. Crossing over**
- D. Transcription**

Q26. STR analysis uses which technique for fragment separation?

- A. Chromatography**
- B. Capillary electrophoresis**
- C. Centrifugation**
- D. Spectroscopy**

Q27. Which blood group is universal donor? [CUET 2023 PYQ]

- A. AB⁺**
- B. O⁺**
- C. O⁻**
- D. AB⁻**

Q28. Which hormone regulates ovulation? [NEET 2020 PYQ]

- A. FSH**
- B. LH**
- C. Estrogen**
- D. Progesterone**



Q29. Which antibody dominates secondary immune response? [NEET 2019 PYQ]

- A. IgM**
- B. IgA**
- C. IgE**
- D. IgG**

Q30. Which nitrogenous base is absent in RNA? [NEET 2018 PYQ]

- A. Adenine**
- B. Guanine**
- C. Cytosine**
- D. Thymine**

Q31. Which forensic evidence is MOST individual-specific? [CUET 2023 PYQ]

- A. Blood group**
- B. Hair colour**
- C. DNA profile**
- D. Height**

Q32. Which enzyme synthesizes cDNA? [NEET 2019 PYQ]

- A. DNA polymerase**
- B. RNA polymerase**
- C. Reverse transcriptase**
- D. Ligase**

Q33. Which technique increases homozygosity fastest? [NEET 2022 PYQ]

- A. Outbreeding**
- B. Inbreeding**
- C. Hybridization**
- D. Mutation breeding**



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Q34. Which structure indicates common ancestry? [NEET 2020 PYQ]

- A. Analogous**
- B. Vestigial**
- C. Homologous**
- D. Adaptive**

Q35. CO₂ is transported in blood mainly as: [NEET 2023 PYQ]

- A. Carbamino-Hb**
- B. Dissolved CO₂**
- C. Bicarbonate ions**
- D. Carbonic acid**

Q36. Which immunoglobulin is produced first in primary immune response? [NEET 2021 PYQ]

- A. IgG**
- B. IgA**
- C. IgE**
- D. IgM**

Q37. Which type of DNA is maternally inherited?

- A. Autosomal**
- B. Nuclear**
- C. Mitochondrial**
- D. Y-linked**

Q38. Which structure has Haversian canals?

- A. Compact bone**
- B. Spongy bone**
- C. Cartilage**
- D. Tendon**



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Q39. Which enzyme joins Okazaki fragments?

- A. Helicase**
- B. Ligase**
- C. Gyrase**
- D. Primase**

Q40. Which population parameter disturbs Hardy–Weinberg equilibrium? [NEET 2021 PYQ]

- A. Large population**
- B. Random mating**
- C. Migration**
- D. No mutation**

Q41. Which blood group is universal recipient?

- A. IAIA**
- B. IBIB**
- C. IAIB**
- D. ii**

Q42. Which hormone delays senescence in plants? [NEET 2021 PYQ]

- A. Auxin**
- B. Gibberellin**
- C. Cytokinin**
- D. Ethylene**

Q43. Which is NOT an ex-situ conservation method? [NEET 2019 PYQ]

- A. Zoo**
- B. Seed bank**
- C. Wildlife sanctuary**
- D. Cryopreservation**



Q44. Which chromosome abnormality causes Cri-du-chat syndrome?

- A. Trisomy 21**
- B. Deletion of 5p**
- C. Monosomy X**
- D. Trisomy 18**

Q45. Which tissue is MOST stable for long-term forensic evidence?

- A. Muscle**
- B. Liver**
- C. Brain**
- D. Bone**

Q46. Which organ generates pacemaker impulse?

- A. AV node**
- B. SA node**
- C. Purkinje fibres**
- D. Bundle of His**

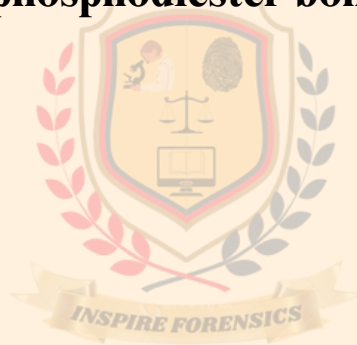
Q47. Which biodiversity hotspot shows maximum species richness?

[CUET 2022 PYQ]

- A. Taiga**
- B. Temperate Forest**
- C. Tropical rainforest**
- D. Tundra**

Q48. Which enzyme forms phosphodiester bond?

- A. Ligase**
- B. Helicase**
- C. Primase**
- D. Gyrase**



Q49. Opening and closing of stomata is controlled by: [NEET 2020 PYQ]

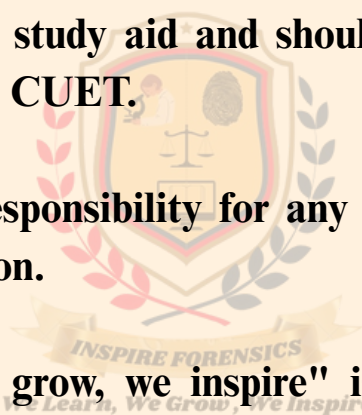
- A. Na^+**
- B. K^+**
- C. Ca^{2+}**
- D. Mg^{2+}**

Q50. DNA is ideal forensic evidence because it is: [CUET 2023 PYQ]

- A. Easily altered**
- B. Same in all humans**
- C. Highly polymorphic**
- D. Environment dependent**

Disclaimer

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Answer Key

Question number	Answer Key		Question number	Answer Key
1	C		19	C
2	C		20	D
3	C		21	C
4	B		22	C
5	D		23	C
6	B		24	C
7	D		25	C
8	C		26	B
9	B		27	C
10	B		28	B
11	C		29	D
12	C		30	D
13	B		31	C
14	C		32	C
15	C		33	B
16	A		34	C
17	C		35	C
18	B		36	D

Answer Key

Question number	Answer Key
37	C
38	A
39	B
40	C
41	C
42	C
43	C
44	B
45	D
46	B
47	C
48	A
49	C
50	B